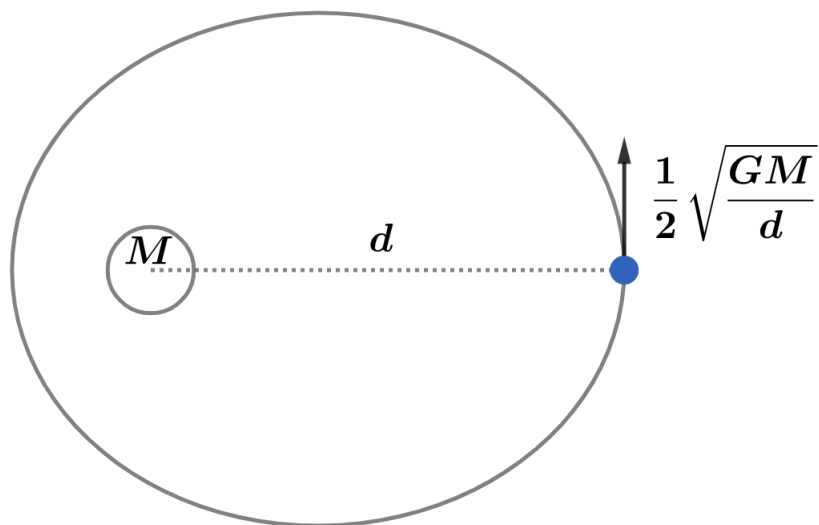


2022A F=ma Exam: Problem 24

Kevin S. Huang



Recall from the elliptical orbit equations, the apogee distance is given by

$$r_1 = a + c$$

The speed at the apogee is given by

$$v_1 = \frac{a - c}{b} \sqrt{\frac{GM}{a}}$$

where a and b are the semi-major and semi-minor axes of the ellipse. In our case,

$$\begin{aligned} d &= a + c \\ \frac{1}{2} \sqrt{\frac{GM}{d}} &= \frac{a - c}{b} \sqrt{\frac{GM}{a}} \end{aligned}$$

We want to find the area of the orbit, $A = \pi ab$. Multiplying the two equations,

$$\begin{aligned} \frac{\sqrt{GMd}}{2} &= \frac{a^2 - c^2}{b} \sqrt{\frac{GM}{a}} \\ \frac{\sqrt{GM(a + c)}}{2} &= b \sqrt{\frac{GM}{a}} = \sqrt{(a + c)(a - c)} \sqrt{\frac{GM}{a}} \end{aligned}$$

using $b^2 = a^2 - c^2$ for an ellipse. Simplifying,

$$\frac{1}{4} = \frac{a - c}{a}$$

$$c = \frac{3a}{4}$$

We have

$$d = a + c = a + \frac{3a}{4} = \frac{7a}{4}$$

Thus,

$$a = \frac{4d}{7}, c = \frac{3d}{7}, b = \sqrt{a^2 - c^2} = \frac{\sqrt{7}}{7}d$$

so the area is

$$A = \pi ab = \pi \left(\frac{4d}{7} \right) \left(\frac{\sqrt{7}}{7}d \right) = \frac{4}{7} \frac{1}{\sqrt{7}} \pi d^2$$

and the answer is B.